

Math 105: Calculus with Algebra
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quiz archive

Gradescope code: **B5J8EN**

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- (1) Solve the equation $|x - 2| = 7$ for x .

Solution. Since $|x - 2| = 7$, either $x - 2 = 7$ or $x - 2 = -7$. Thus $x = 9$ or $x = -5$.

- (2) Solve the inequality $|x + 3| < 4$.

Solution. The inequality $|x + 3| < 4$ is equivalent to $-4 < x + 3 < 4$. Subtracting 3 throughout gives $-7 < x < 1$.

Thus the solution is $(-7, 1)$. On the number line, shade the interval between -7 and 1 , with open circles at -7 and 1 .

- (3) Express

$$1 + \frac{1}{\frac{x}{x+2}}$$

as a single fraction.

Solution. We have

$$1 + \frac{1}{\frac{x}{x+2}} = 1 + \frac{x+2}{x} = \frac{x+x+2}{x} = \frac{2x+2}{x}.$$

Thus the single fraction is $\frac{2x+2}{x}$.

The original expression is defined only when $x \neq -2$ and $x \neq 0$.

(1) Consider the line

$$x + 2y = 2.$$

(a) To find the slope, solve the equation for y :

$$x + 2y = 2$$

$$2y = -x + 2$$

$$y = -\frac{1}{2}x + 1.$$

Thus the slope is

$$\boxed{-\frac{1}{2}}.$$

(b) A line parallel to this line has the same slope, so its slope is $-\frac{1}{2}$. Since it passes through $(2, -1)$, we can use point-slope form:

$$y - (-1) = -\frac{1}{2}(x - 2).$$

Thus

$$y + 1 = -\frac{1}{2}x + 1,$$

and hence

$$\boxed{y = -\frac{1}{2}x}.$$

(c) A perpendicular line has slope equal to the negative reciprocal of the original slope. Since the original slope is $-\frac{1}{2}$, the perpendicular slope is

$$\boxed{2}.$$

(2) Let

$$f(x) = 3x^2 + 2.$$

We want to compute

$$\frac{f(a+h) - f(a)}{h}.$$

First,

$$f(a+h) = 3(a+h)^2 + 2.$$

Expanding gives

$$f(a+h) = 3(a^2 + 2ah + h^2) + 2 = 3a^2 + 6ah + 3h^2 + 2.$$

Also,

$$f(a) = 3a^2 + 2.$$

Therefore

$$\begin{aligned} \frac{f(a+h) - f(a)}{h} &= \frac{3a^2 + 6ah + 3h^2 + 2 - (3a^2 + 2)}{h} \\ &= \frac{6ah + 3h^2}{h} \\ &= \boxed{6a + 3h}. \end{aligned}$$

(3) Let

$$g(x) = \begin{cases} x^2, & \text{if } x \leq 4, \\ -|x-4|, & \text{if } x > 4. \end{cases}$$

(a) To compute $g(0)$, notice that $0 \leq 4$, so we use the first part of the definition:

$$g(0) = 0^2 = \boxed{0}.$$

Since $5 > 4$, we use the second part to compute $g(5)$:

$$g(5) = -|5 - 4| = -|1| = \boxed{-1}.$$

(b) For $x \leq 4$, the graph is the parabola

$$y = x^2.$$

Since $x = 4$ is included, there is a closed point at

$$(4, 16).$$

For $x > 4$, we have $x - 4 > 0$, so

$$|x - 4| = x - 4.$$

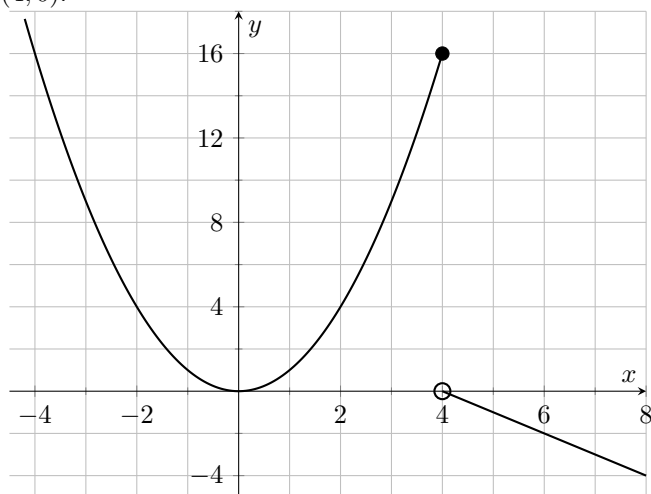
Therefore

$$-|x - 4| = -(x - 4) = 4 - x.$$

Thus the second piece is the line

$$y = 4 - x, \quad x > 4,$$

with an open point at $(4, 0)$.



The first piece covers all $x \leq 4$ and the second covers all $x > 4$. Therefore the domain is

$$\boxed{(-\infty, \infty)}.$$

The parabola gives every $y \geq 0$, while the line gives every $y < 0$. Together these give every real number.

Therefore the range is

$$\boxed{(-\infty, \infty)}.$$

- (1) The graph of $y = f(x)$ is shown below, and we want to sketch

$$y = f(x - 1) + 2.$$

The expression $f(x - 1)$ shifts the graph of f **one unit to the right**. Then adding 2 shifts the graph **two units up**.

Thus every point

$$(x, y)$$

on the original graph moves to

$$(x + 1, y + 2).$$

In particular,

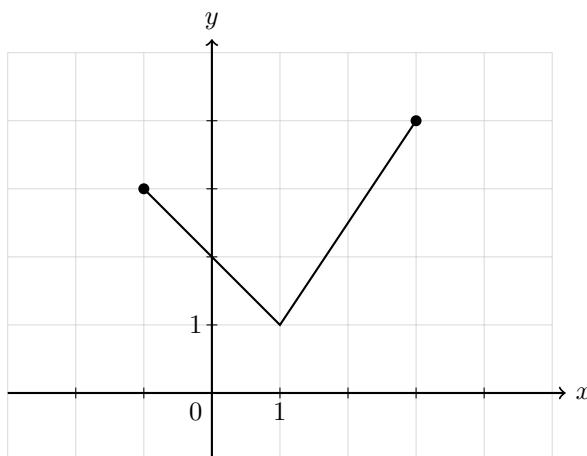
$$(-2, 1) \mapsto (-1, 3),$$

$$(0, -1) \mapsto (1, 1),$$

and

$$(2, 2) \mapsto (3, 4).$$

Therefore the transformed graph is:



Shift right 1 unit and up 2 units.

- (2) Let

$$f(x) = \sqrt{x+1} \quad \text{and} \quad g(x) = x^2 - 2.$$

We want to compute

$$(f \circ g)(2).$$

Recall that

$$(f \circ g)(2) = f(g(2)).$$

First compute the inside function:

$$g(2) = 2^2 - 2 = 4 - 2 = 2.$$

Then

$$f(g(2)) = f(2) = \sqrt{2+1} = \sqrt{3}.$$

Therefore

$$(f \circ g)(2) = \sqrt{3}.$$

(3) Find

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}.$$

If we substitute $x = 3$ immediately, we get

$$\frac{3^2 - 9}{3 - 3} = \frac{0}{0},$$

so we need to simplify first.

Factor the numerator:

$$x^2 - 9 = (x - 3)(x + 3).$$

Therefore, for $x \neq 3$,

$$\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3.$$

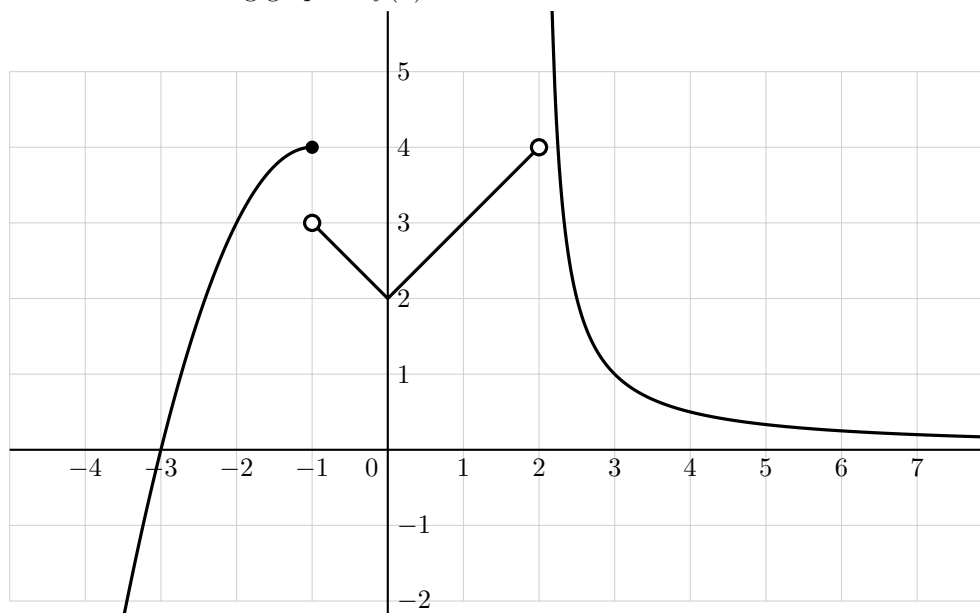
Now we can take the limit:

$$\lim_{x \rightarrow 3} (x + 3) = 3 + 3 = 6.$$

Thus

$$\boxed{\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6}.$$

- (1) (5 points) Consider the following graph of $f(x)$.



- (a) As x approaches 2 from the right, the graph rises without bound. Therefore

$$\lim_{x \rightarrow 2^+} f(x) = +\infty.$$

- (b) As x approaches 2 from the left, the graph approaches the open point $(2, 4)$. Therefore

$$\lim_{x \rightarrow 2^-} f(x) = 4.$$

- (c) The right-hand limit is $+\infty$, while the left-hand limit is 4. Since the one-sided limits do not agree,

$$\lim_{x \rightarrow 2} f(x) \text{ does not exist.}$$

- (d) As $x \rightarrow -1^-$, the graph approaches 4, while as $x \rightarrow -1^+$, the graph approaches 3. Since the one-sided limits are different,

$$\lim_{x \rightarrow -1} f(x) \text{ does not exist.}$$

- (e) Near $x = 0$, the graph lies on the two line segments meeting at $(0, 2)$, so from both sides $f(x)$ approaches 2. Thus

$$\lim_{x \rightarrow 0} f(x) = 2.$$

(2) (1 point)

$$\lim_{x \rightarrow -3} \frac{x+3}{x^2+x+1}.$$

The denominator is nonzero at $x = -3$, since $(-3)^2 + (-3) + 1 = 7$. Therefore direct substitution gives

$$\boxed{0}.$$

(3) (1 point)

$$\lim_{x \rightarrow 3} \frac{x-7}{3-x}.$$

As $x \rightarrow 3$, the numerator approaches -4 . If $x \rightarrow 3^-$, then $3-x \rightarrow 0^+$, so the quotient approaches $-\infty$. If $x \rightarrow 3^+$, then $3-x \rightarrow 0^-$, so the quotient approaches $+\infty$. Therefore

the limit does not exist.

(4) (1 point)

$$\lim_{x \rightarrow -3} \frac{x+3}{x^2-2x-15}.$$

Factor the denominator:

$$x^2 - 2x - 15 = (x-5)(x+3).$$

For $x \neq -3$,

$$\frac{x+3}{(x-5)(x+3)} = \frac{1}{x-5}.$$

Therefore

$$\boxed{-\frac{1}{8}}.$$

(5) (2 points)

$$\lim_{x \rightarrow -3} \frac{\frac{1}{1-x} - \frac{1}{4}}{x+3}.$$

Combine the fractions in the numerator:

$$\frac{1}{1-x} - \frac{1}{4} = \frac{4 - (1-x)}{4(1-x)} = \frac{x+3}{4(1-x)}.$$

Hence, for $x \neq -3$,

$$\frac{\frac{x+3}{4(1-x)}}{x+3} = \frac{1}{4(1-x)}.$$

Now substitute $x = -3$:

$$\boxed{\frac{1}{16}}.$$

Math 105: Midterm 1 Solution Guide

1. Consider the line L_1 given by $3x + 4y = 5$.

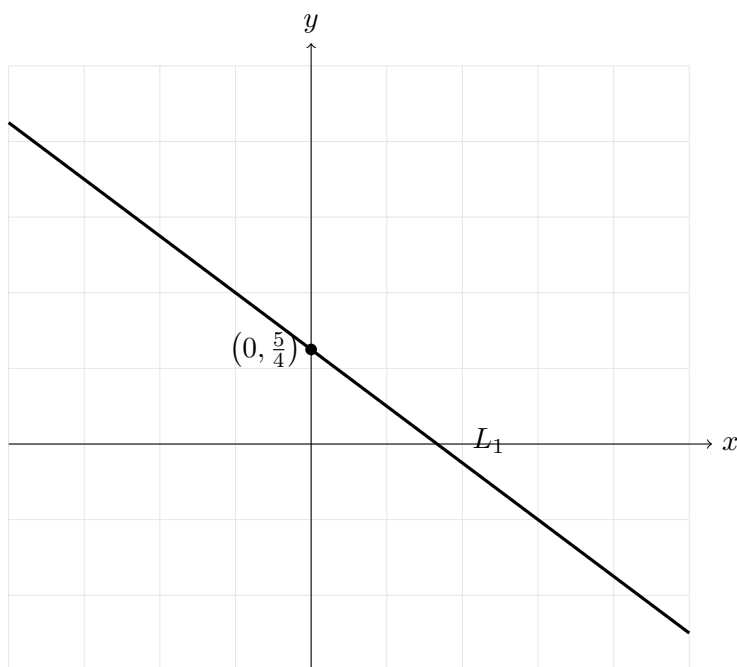
(a) Solve for y :

$$3x + 4y = 5 \quad \implies \quad y = -\frac{3}{4}x + \frac{5}{4}.$$

Thus L_1 has slope $-\frac{3}{4}$ and y -intercept

$$\left(0, \frac{5}{4}\right).$$

A convenient second point is the x -intercept $(\frac{5}{3}, 0)$.



(b) A line perpendicular to L_1 has slope equal to the opposite reciprocal of $-\frac{3}{4}$, namely $\frac{4}{3}$. Using the point $(-5, -1)$,

$$y + 1 = \frac{4}{3}(x + 5).$$

Solving for y gives

$$y = \frac{4}{3}x + \frac{17}{3}.$$

(c) A line parallel to L_1 must have the same slope. Therefore

$$m = -\frac{3}{4}.$$

2. Let $f(x) = x^2 - x$ and $g(x) = \sqrt{x + 4}$.

(a) Substitute $2x^3$ for x in f :

$$f(2x^3) = (2x^3)^2 - 2x^3 = \boxed{4x^6 - 2x^3}.$$

(b) Substitute x^2 for x in g :

$$g(x^2) = \boxed{\sqrt{x^2 + 4}}.$$

(c)

$$(f \circ g)(x) = f(g(x)) = (\sqrt{x+4})^2 - \sqrt{x+4}.$$

Therefore

$$\boxed{(f \circ g)(x) = x + 4 - \sqrt{x+4}}.$$

The expression is defined when $x + 4 \geq 0$, so its domain is $[-4, \infty)$.

3. Evaluate each limit.

(a)

$$\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x^2 - 3x - 4}.$$

Direct substitution gives $0/0$, so factor:

$$\frac{(x-4)(x+3)}{(x-4)(x+1)} = \frac{x+3}{x+1} \quad (x \neq 4).$$

Thus

$$\boxed{\frac{7}{5}}.$$

(b)

$$\lim_{x \rightarrow -1} \frac{x^2 - 1}{2 - \sqrt{x+5}}.$$

Again direct substitution gives $0/0$. Rationalize the denominator:

$$\frac{x^2 - 1}{2 - \sqrt{x+5}} \cdot \frac{2 + \sqrt{x+5}}{2 + \sqrt{x+5}} = \frac{(x-1)(x+1)(2 + \sqrt{x+5})}{4 - (x+5)}.$$

Since $4 - (x+5) = -(x+1)$, this simplifies to

$$-(x-1)(2 + \sqrt{x+5}).$$

Now substitute $x = -1$:

$$-(-2)(2 + 2) = \boxed{8}.$$

(c)

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{4 - x}.$$

The denominator is nonzero at $x = 2$, so direct substitution works:

$$\frac{4 - 4}{4 - 2} = \boxed{0}.$$

(d)

$$\lim_{x \rightarrow 2} \frac{x+1}{(x-2)^2}.$$

The numerator approaches $3 > 0$, while $(x-2)^2 \rightarrow 0^+$ from both sides. Therefore

$$\boxed{+\infty}.$$

(e)

$$\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{|x - 1|}.$$

Factor the numerator:

$$\frac{(x-1)(x-2)}{|x-1|}.$$

For $x > 1$, $|x-1| = x-1$, so

$$\lim_{x \rightarrow 1^+} \frac{x^2 - 3x + 2}{|x-1|} = \lim_{x \rightarrow 1^+} (x-2) = -1.$$

For $x < 1$, $|x-1| = -(x-1)$, so

$$\lim_{x \rightarrow 1^-} \frac{x^2 - 3x + 2}{|x-1|} = \lim_{x \rightarrow 1^-} -(x-2) = 1.$$

The one-sided limits are different, so

$$\boxed{\text{DNE}}.$$

4. Sketch $f(x) = 1 + \frac{2}{x-1}$.

The denominator is zero at $x = 1$, so the vertical asymptote is

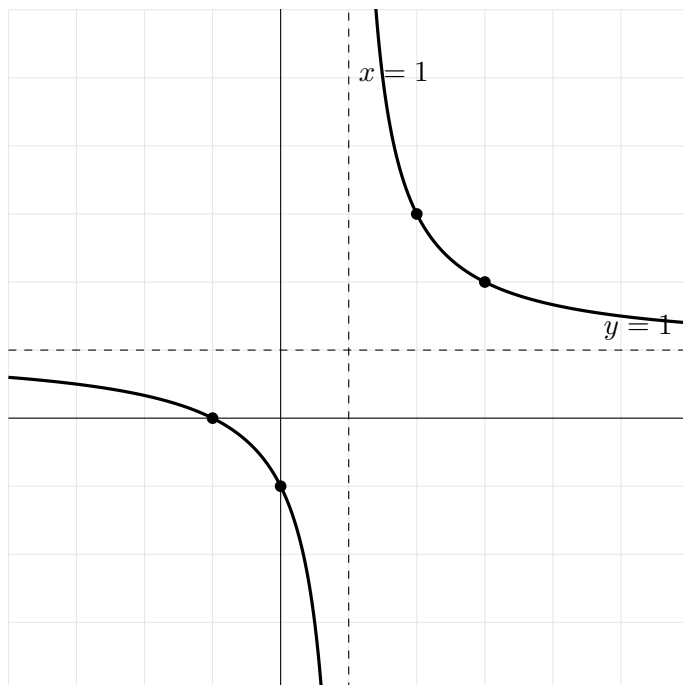
$$\boxed{x = 1}.$$

As x becomes very large in magnitude, $\frac{2}{x-1} \rightarrow 0$, so the horizontal asymptote is

$$\boxed{y = 1}.$$

Some convenient points are

$$f(-1) = 0, \quad f(0) = -1, \quad f(2) = 3, \quad f(3) = 2.$$



5. Let $f(x) = \frac{1}{x}$.

(a)

$$\frac{f(x+h) - f(x)}{h} = \frac{\frac{1}{x+h} - \frac{1}{x}}{h}.$$

Combine the fractions in the numerator:

$$\frac{1}{x+h} - \frac{1}{x} = \frac{x - (x+h)}{x(x+h)} = \frac{-h}{x(x+h)}.$$

Therefore

$$\frac{f(x+h) - f(x)}{h} = \boxed{-\frac{1}{x(x+h)}}.$$

(b) Using the simplified expression from part (a),

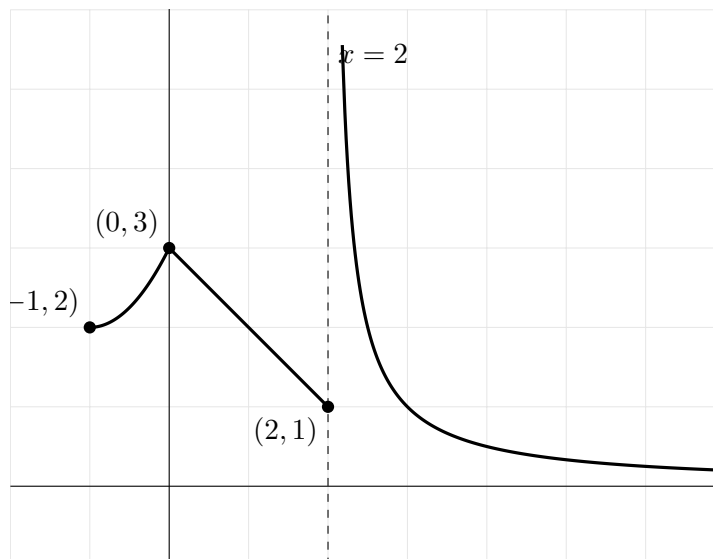
$$\lim_{h \rightarrow 0} -\frac{1}{x(x+h)} = \boxed{-\frac{1}{x^2}},$$

for $x \neq 0$.

6. Consider

$$f(x) = \begin{cases} \frac{1}{x-2}, & x > 2, \\ 3 - |x|, & 0 < x \leq 2, \\ (x+1)^2 + 2, & -1 \leq x \leq 0. \end{cases}$$

(a) Since $0 < x \leq 2$ in the middle piece, $|x| = x$, so that piece is the line $y = 3 - x$. A sketch is shown below.



(b) Since the middle piece includes $x = 2$,

$$f(2) = 3 - |2| = \boxed{1}.$$

(c) The three pieces cover every input from -1 onward, so

$$\boxed{\text{Domain} = [-1, \infty)}.$$

For $x > 2$, the function $1/(x - 2)$ takes every positive value. Therefore

$$\boxed{\text{Range} = (0, \infty)}.$$

(d) At $x = 0$,

$$\lim_{x \rightarrow 0^-} f(x) = 3 \quad \text{and} \quad \lim_{x \rightarrow 0^+} f(x) = 3.$$

Hence

$$\boxed{\lim_{x \rightarrow 0} f(x) = 3}.$$

At $x = 2$,

$$\lim_{x \rightarrow 2^-} f(x) = 1, \quad \lim_{x \rightarrow 2^+} f(x) = +\infty.$$

The one-sided limits do not agree, so

$$\boxed{\lim_{x \rightarrow 2} f(x) = \text{DNE}}.$$